

Weather Forecast Variables & Metrics - Quick Summary

Why zonal = west-east and meridional = north-south?

A **zone** of latitude runs around Earth parallel to the equator, so moving along a latitude circle is mainly **west-east**. Hence **zonal wind** is the west-east component, usually written **u**.

A **meridian** is a line of longitude joining the North and South Poles, so moving along a meridian is **north-south**. Hence **meridional wind** is the north-south component, usually written **v**.

Memory trick: **zonal** -> **zones/latitude bands** -> **W-E**; **meridional** -> **meridians/longitude lines** -> **N-S**.

1. Common forecast variables

Variable	Notation	Meaning / intuition
2 m temperature	T2m	Near-surface air temperature at about human height.
10 m zonal wind	U10	West-east wind component at 10 m. Positive u is eastward.
10 m meridional wind	V10	North-south wind component at 10 m. Positive v is northward.
Mean sea-level pressure	MSLP	Pressure adjusted to sea level; highlights highs, lows and cyclones.
Precipitation	TP / precip.	Rain/snow accumulated over a specified time window.
Geopotential / height	Z, e.g. Z500	Height/energy structure of pressure surfaces; tracks large-scale ridges and troughs.
Temperature aloft	T, e.g. T850	Temperature on a pressure surface; useful for air-mass structure.
Specific humidity	Q, e.g. Q700	Mass fraction of water vapour; indicates atmospheric moisture.
Zonal wind aloft	U, e.g. U250	West-east flow at a pressure level; U250 captures jet-stream dynamics.
Meridional wind aloft	V	North-south flow at a pressure level.
Pressure vertical velocity	omega	dp/dt; negative typically means rising air because pressure falls upward.

Pressure-level notation: the number is pressure, not altitude. For example, Z500 means geopotential/geopotential height at 500 hPa; Q700 means specific humidity at 700 hPa. Roughly: 850 hPa ~ 1.5 km, 700 hPa ~ 3 km, 500 hPa ~ 5-6 km, 250 hPa ~ 10-11 km, though the exact altitude varies.

Representative four used in AI-weather analysis: Z500 (large-scale circulation), T2m (surface weather), U250 (upper-level/jet flow), Q700 (mid-level moisture).

2. Wind-component example

If $u = 3$ m/s and $v = 4$ m/s, the wind vector is (3, 4) and its speed is $\sqrt{3^2 + 4^2} = 5$ m/s. The u component tells you the east-west contribution; v tells you the north-south contribution.

3. Common evaluation metrics

Metric	Core idea	Interpretation
MAE	$\text{mean}(\text{forecast} - \text{truth})$	Typical absolute error; less sensitive to large outliers than RMSE.
MSE	$\text{mean}((\text{forecast} - \text{truth})^2)$	Squares errors, so large misses are penalized strongly.
RMSE	$\sqrt{\text{MSE}}$	Same units as the variable; common headline deterministic metric.
Latitude-weighted RMSE	RMSE with weights $\sim \cos(\text{latitude})$	Corrects for unequal physical area of latitude-longitude grid cells.
ACC	correlation of forecast and observed anomalies	Measures whether the spatial anomaly pattern is correct, relative to climatology.
Forecast activity	spatial std. dev. of forecast anomalies	Checks whether forecasts are too smooth or too variable.
IE	missing useful anomaly information	How much correctly aligned signal/amplitude is absent.
NE	misaligned/noisy anomaly component	How much forecast variation points in the wrong spatial direction.
Thresholded RMSE	RMSE only where a threshold is exceeded	Focuses verification on extreme conditions rather than all grid points.
Track error / MAE	distance between forecast and observed storm centres	Task-specific metric for cyclone position.

ACC example - why it is different from RMSE

Suppose climatology is [10, 10, 10, 10], reality is [12, 14, 8, 6], so the true anomaly is [+2, +4, -2, -4]. A forecast with anomaly [+1, +2, -1, -2] has exactly the same spatial pattern but only half the amplitude. Its ACC can still be 1, even though its magnitudes are wrong. Therefore ACC measures **pattern agreement**, not full numerical accuracy.

Activity / smoothing example

Truth might contain a strong ridge and deep trough, while an ML forecast places both features correctly but makes them much weaker. ACC may remain high, and RMSE may not look disastrous, but forecast activity drops. This is why activity is useful for detecting overly smooth neural forecasts.

4. A practical metric stack for adversarial robustness

RMSE: how numerically wrong? **ACC:** is the weather pattern right? **Activity:** is the forecast too smooth/wild? **IE/NE:** what useful signal was lost vs replaced by wrong-pattern variation? **Thresholded RMSE:** does robustness hold for extreme weather?

For an adversarial attack, relying on one metric can be misleading. An attack could preserve the broad spatial pattern (high ACC) while suppressing amplitudes (low activity), or create large localized extreme errors that barely affect global RMSE.

Source papers summarized

Vonich & Hakim, AI weather predictability / optimal initial conditions (arXiv:2504.20238; AIES published version); Aardvark Weather (Nature, 2025); Aurora (Nature, 2025). The adversarial-ML survey papers in the original reading list provide general attack/robustness context rather than weather-specific verification definitions.